

Lab 2 - Configuring Basic Single-Area OSPFv2

Answer Sheet

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Part 2: Configure and Verify OSPF Routing

Step 1: Verify OSPF neighbors and routing information.

b.

What command would you use to only see the OSPF routes in the routing table?

`show ip route ospf`

Part 4: Configure OSPF Passive Interfaces

- g. Re-issue the **show ip route** and **show ip ospf neighbor** commands on R1 and R3, and look for a route to the 192.168.2.0/24 network.

What interface is R3 using to route to the 192.168.2.0/24 network? `S0/0/0`

What is the accumulated cost metric for the 192.168.2.0/24 network on R3? `129`

Does R2 show up as an OSPF neighbor on R1? `Yes`

Does R2 show up as an OSPF neighbor on R3? `No`

What does this information tell you?

All data from R3 to the 192.168.2.0/24 network travels through R1. The S0/0/1 interface on R2 is set to passive, which stops it from sharing OSPF routing updates. The total path cost is 129. This is calculated by adding two T1 serial links (64 each) and one Gigabit Ethernet link (1).

- h. Change interface S0/0/1 on R2 to allow it to advertise OSPF routes. Record the commands used below.

`R2(config)# router ospf 1`

`R2(config-router)# no passive-interface s0/0/1`

- i. Re-issue the **show ip route** command on R3.

What interface is R3 using to route to the 192.168.2.0/24 network? `S0/0/1`

What is the accumulated cost metric for the 192.168.2.0/24 network on R3 now and how is this calculated?

`65` (One T1 (1.544 Mb/s) serial link (at a cost of 64) plus the R2 Gigabit 0/0 LAN link (at a cost of 1))

Is R2 listed as an OSPF neighbor to R3? `Yes`

Part 5: Change OSPF Metrics

Step 1: Change the reference bandwidth on the routers.

- a. To reset the reference-bandwidth back to its default value, issue the **auto-cost reference-bandwidth 100** command on all three routers.

```
R1(config)# router ospf 1
```

```
R1(config-router)# auto-cost reference-bandwidth 100
```

```
% OSPF: Reference bandwidth is changed.
```

```
Please ensure reference bandwidth is consistent across all routers.
```

Why would you want to change the OSPF default reference-bandwidth?

Current Issue: Modern networking equipment now operates at speeds much higher than 100 Mb/s. To ensure routing costs remain accurate for these faster links, the default reference-bandwidth must be increased. A higher reference value allows the system to correctly differentiate between high-speed interfaces like Gigabit or 10-Gigabit Ethernet.

Step 2: Change the bandwidth for an interface.

g.

Explain how the costs to the 192.168.3.0/24 and 192.168.23.0/30 networks from R1 were calculated.

Path to 192.168.3.0/24: The total cost is 782, which includes the R1 S0/0/1 serial link (781) and the R3 G0/0 interface (1). Path to 192.168.23.0/30: The total cost is 845, calculated by adding the R1 S0/0/1 link (781) and the R3 S0/0/1 serial link (64).

- i. Issue the **bandwidth 128** command on all remaining serial interfaces in the topology.

What is the new accumulated cost to the 192.168.23.0/24 network on R1? Why?

Total Cost: The final cost to reach the 192.168.23.0/24 network is 1,562. Since each serial link now costs 781, and the traffic must pass through two serial links, the total is calculated as $781 + 781 = 1,562$.

Step 3: Change the route cost.

c.

Explain why the route to the 192.168.3.0/24 network on R1 is now going through R2?

OSPF always selects the path with the lowest accumulated cost. The route through R1-S0/0/0, R2-S0/0/1, and R3-G0/0 is chosen because its total cost is 1,563 ($781 + 781 + 1$). The direct route via R1-S0/0/1 and R3-G0/0 is rejected because its cost is higher at 1,566 ($1,565 + 1$).

Reflection

1. Why is it important to control the router ID assignment when using the OSPF protocol?

The Router ID (RID) is the primary factor in choosing the Designated Router and Backup Designated Router. If the RID is tied to a physical interface, the ID can change if that interface fails, causing network instability. To ensure a constant RID, it should be manually set using the router-id command or assigned to a loopback interface, as loopback interfaces never go down.

2. Why is the DR/BDR election process not a concern in this lab?

The DR/BDR election only occurs on multi-access networks like Ethernet, where many devices share the same segment. Since the serial links in this lab are point-to-point connections between only two routers, no election is performed. OSPF naturally forms a direct adjacency without needing a designated controller on these specific links.

3. Why would you want to set an OSPF interface to passive?

Configuring a LAN interface as passive stops the router from sending unnecessary OSPF updates to local devices, which saves bandwidth and CPU resources. Even though the interface is silent, the router still advertises that LAN network to all its neighbors, ensuring the rest of the network knows how to reach it.

4. What aspect of OSPF makes the protocol hierarchical and permits the creation of very scalable networks?

The Area-based design is the aspect of OSPF that makes it hierarchical and highly scalable. By dividing a large network into different Areas (with Area 0 as the Backbone), OSPF limits the amount of routing information each router needs to process. This design reduces CPU load and prevents a single link failure from causing a recalculation across the entire global network.

5. What single OSPF router configuration command allows the assignment of OSPF area 0 to all interfaces in the range 10.0.0.0 to 10.255.255.255?

`network 10.0.0.0 0.255.255.255 area 0`

6. A router has three routes to the same network: one route from a RIP with a metric of 4, another from OSPF with a metric of 3053092, and another from EIGRP with a metric of 4039043. Which of the three routes will be used for the routing decision?

The router will use the EIGRP route.

Even though EIGRP has the highest metric number, the router first compares the Administrative Distance (AD) of each protocol. Since EIGRP has the lowest AD (90), it is considered the most "trustworthy" source compared to OSPF (110) or RIP (120). The metric is only compared if the AD is the same.